# P.6 Trigonometric Functions

Name _____ Period ____ Date _____

AP Calculus P.6 · Student sheet

**I can…**

- find exact values of the trigonometric functions in radians.
- use the period and the even-odd properties to evaluate trigonometric functions.
- find the amplitude and period of a sinusoidal function and write its equation.

**Vocabulary** · fill in the missing word

**Periodic function** — a function f with a _____ number p so that f(x + p) = f(x) for every x in its domain

**Period** — the _____ such number p

**Amplitude** — |A| for y = _____ = A cos(ωx)

## Part 1 · Sine from the unit circle

Use the unit circle.

| x | 0 | π/6 | π/2 | 5π/6 | π | 7π/6 | 3π/2 | 11π/6 | 2π |
|---|---|---|---|---|---|---|---|---|---|
| sin x | | | | | | | | | |
| cos x | | | | | | | | | |

## Part 2 · Notes and examples (fill in each note, then work the example)

**Domain and range:** sine and cosine are defined for all real numbers and have range [−1, 1]; tan x is undefined at odd integer multiples of _____.

### Example 1

Find the exact value.

a. $\sin(\frac{5\pi}{6})$

b. $\cos(\frac{7\pi}{6})$

c. $\tan(\frac{3\pi}{4})$

**Periods:** sine, cosine, secant, and cosecant have period 2π; tangent and cotangent have period _____.

### Example 2

Find the exact value.

a. $\tan(\frac{\pi}{4} + 2\pi)$

b. $\sin(\frac{17\pi}{6})$

**Even-odd:** cosine and secant are even; sine, tangent, cosecant, and cotangent are _____.

### Example 3

Use the even-odd properties to find the exact value.

a. $\tan(-\frac{\pi}{4})$

b. $\sin(-\frac{3\pi}{2})$

c. $\csc(-\frac{\pi}{3})$

**Sinusoids:** $y = A\sin(\omega x)$ and $y = A\cos(\omega x)$, $\omega > 0$, have amplitude $|A|$ and period _____.

### Example 4

Find the amplitude and period.

a. $g(x) = (\frac{1}{2})\cos(\pi x)$

b. $f(x) = -2\cos((\frac{3}{2})x)$

### Example 5

a. Write a sine function with amplitude 2 and period $\pi$.

b. Write a cosine function with amplitude 3 and period $4\pi$.

## Part 3 · Partner practice

### Partner 1

Find the amplitude, period, and range of $f(x) = 3\cos(2x) - 4$.

### Partner 2

Find $\sin(-\frac{\pi}{2})$ and $\cos(\frac{13\pi}{6})$.